ECON 2050 – Mathematical Economics

Optimization III: From Linear Programming to Nonlinear Programming

Complementary slackness

Theorem

If $\mathbf{x}^*$ is a solution to the primal and $\mathbf{y}^*$ is a solution to the dual. Then

1. $y_i^* > 0$ *implies* $\sum_j a_{ij} x_j^* = b_i$;
2. $\sum_{j=1}^n a_{ij} x_j^* < b_i$ *implies* $y_i^* = 0$;
3. $x_j^* > 0$ *implies* $\sum_{i=1}^m a_{ij} y_i^* = c_j$;
4. $\sum_{i=1}^m a_{ij} y_i^* > c_j$ *implies* $x_j^* = 0$.

Interpretation:

1. $y_i^* > 0$ means that the marginal value of resource i is strictly positive, and therefore at the optimum all the available stock of resource i must be used.
2. $\sum_{j=1}^n a_{ij} x_j^* < b_i$ means that not all available stock of resource i is used, then it must be that the marginal value of resource i, y_i^*, is zero.

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If $\mathbf{x}^*$ is a solution to the primal and $\mathbf{y}^*$ is a solution to the dual. Then

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2. $\sum_{j=1}^n a_{ij} x_j^* < b_i$ *implies* $y_i^* = 0$;
3. $x_j^* > 0$ *implies* $\sum_{i=1}^m a_{ij} y_i^* = c_j$;
4. $\sum_{i=1}^m a_{ij} y_i^* > c_j$ *implies* $x_j^* = 0$.

Interpretation (Continued): Recall from the previous lecture that c_j is the marginal benefit from running activity j, and the imputed marginal cost of running activity j is $\sum_{i=1}^m a_{ij} y_i$. Then

3. $x_j^* > 0$ means that the (imputed) marginal cost and marginal benefit must equal, as otherwise x_j should be set to 0.
4. $\sum_{i=1}^m a_{ij} y_i^* > c_j$ means that the (imputed) marginal cost is strictly higher than marginal benefit, so engaging in activity j would incur a loss, hence x_j^* must be zero.

Consequently, we can treat every primal constraint and its corresponding dual variable as a "complementary pair:" if one is slack the other one must be binding. Thus,

$$y_i^* \left(\sum_{j=1}^{n} a_{ij} x_j^* - b_i \right) = 0 \quad \text{for every } i = 1, \ldots, m. \tag{CS1}$$

Similarly, we can treat every dual constraint and its corresponding primal variable as a "complementary pair:" if one is slack the other one must be binding. Thus,

$$x_j^* \left(\sum_{i=1}^{m} a_{ij} y_i^* - c_j \right) = 0 \quad \text{for every } j = 1, \ldots, n. \tag{CS2}$$

The conditions (CS1) and (CS2) are called the **complementary slackness conditions**.

- As we will soon see, these conditions can be very useful in solving linear or even non-linear programming problems.

Nonlinear programming

Although linear programs are ubiquitous in economics, finance, and business, linearity of objective or constraint functions may fail in many important problems. For example:

- In a consumer's utility maximization problem, the utility function u may not be linear:

$$\max_{(x,y)\in\mathbb{R}^2} u(x,y) \qquad \text{s.t.} \quad p_x x + p_y y \leq M$$
$$x \geq 0,\ y \geq 0$$

- A firm whose production function is f, which may not be linear, wants to minimize costs $rK + wL$ while producing at least b units of output (r and w are the unit costs of the two input factors x and y, respectively):

$$\min_{(K,L)\in\mathbb{R}^2} rK + wL \qquad \text{s.t.} \quad f(K,L) \geq b$$
$$K \geq 0,\ L \geq 0$$

Therefore, we need to extend linear programming techniques to settings with non-linear objective function or constraint functions.

In what follows we define a more general problem of optimizing a (potentially non-linear) function subject to a few (potentially non-linear) constraints, called **nonlinear programming**.

- Let S be an open subset of $\mathbb{R}^n$.
- The objective function, $f : S \to \mathbb{R}$, is continuously differentiable.
- There are m constraint functions; for each $i = 1, 2, \ldots, m$, $g_i : S \to \mathbb{R}$ is continuously differentiable.
- We consider the maximization problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{subject to} \quad g_i(\boldsymbol{x}) \leq 0 \text{ for all } i = 1, \ldots, m$$

Any non-linear program can be transformed into the above form; however, unlike linear programs, there are pitfalls for the careless (more on that later).

Suppose $\boldsymbol{x}^* \in S$ is a solution to the non-linear program

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{subject to} \quad g_i(\boldsymbol{x}) \leq 0 \text{ for all } i = 1, \ldots, m;$$

denote the constraint set by

$$C = \{\boldsymbol{x} \in S : g_i(\boldsymbol{x}) \leq 0 \text{ for all } i = 1, \ldots, m\}.$$

Then for any $\boldsymbol{x} \in C$, $f(\boldsymbol{x}^*) \geq f(\boldsymbol{x})$.

Because f is continuously differentiable, around $\boldsymbol{x}^*$ we can approximate f by its (first-order) Taylor approximation (see lecture 3 slide 24):

$$f(\boldsymbol{x}^* + \boldsymbol{h}) \approx f(\boldsymbol{x}^*) + \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{h}.$$

Similarly, we can approximate each of the constraint functions g_i around $\boldsymbol{x}^*$:

$$g_i(\boldsymbol{x}^* + \boldsymbol{h}) \approx g_i(\boldsymbol{x}^*) + \nabla g_i(\boldsymbol{x}^*) \cdot \boldsymbol{h}.$$

Because $\boldsymbol{x}^*$ maximizes f on C, the following maximization problem must have a solution $\boldsymbol{h} = 0$:

$$\max_{\boldsymbol{h}} f(\boldsymbol{x}^*) + \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{h} \qquad \text{subject to} \quad g_i(\boldsymbol{x}^*) + \nabla g_i(\boldsymbol{x}^*) \cdot \boldsymbol{h} \leq 0 \text{ for all } i = 1, \ldots, m.$$

Note that dropping the constant terms of the objective functions doesn't affect the solutions, and therefore the problem above is equivalent to the following *linear* program:

$$\max_{\boldsymbol{h}} \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{h} \qquad \text{subject to} \quad \nabla g_i(\boldsymbol{x}^*) \cdot \boldsymbol{h} \leq -g_i(\boldsymbol{x}^*) \text{ for all } i = 1, \ldots, m.$$

If this still doesn't seem familiar to you, let $\boldsymbol{c} = \nabla f(\boldsymbol{x}^*)$,

$$\boldsymbol{A} = \begin{pmatrix} \nabla g_1(\boldsymbol{x}^*) \\ \nabla g_2(\boldsymbol{x}^*) \\ \vdots \\ \nabla g_m(\boldsymbol{x}^*) \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} -g_1(\boldsymbol{x}^*) \\ -g_2(\boldsymbol{x}^*) \\ \vdots \\ -g_m(\boldsymbol{x}^*) \end{pmatrix},$$

the problem can be written as $\max \boldsymbol{c} \cdot \boldsymbol{h}$ subject to $\boldsymbol{A}\boldsymbol{h} \leq \boldsymbol{b}$.

The dual of the problem $\max \boldsymbol{c} \cdot \boldsymbol{h}$ subject to $\boldsymbol{A}\boldsymbol{h} \leq \boldsymbol{b}$ is

$$\min \boldsymbol{\lambda} \cdot \boldsymbol{b} \qquad \text{subject to} \quad \boldsymbol{\lambda}\boldsymbol{A} = \boldsymbol{c} \text{ and } \boldsymbol{\lambda} \geq 0.$$

Recalling that $\boldsymbol{c} = \nabla f(\boldsymbol{x}^*)$,

$$\boldsymbol{A} = \begin{pmatrix} \nabla g_1(\boldsymbol{x}^*) \\ \nabla g_2(\boldsymbol{x}^*) \\ \vdots \\ \nabla g_m(\boldsymbol{x}^*) \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} -g_1(\boldsymbol{x}^*) \\ -g_2(\boldsymbol{x}^*) \\ \vdots \\ -g_m(\boldsymbol{x}^*) \end{pmatrix},$$

the dual can be written as

$$\begin{aligned} \min_{\boldsymbol{\lambda}} \; & -\lambda_1 g_1(\boldsymbol{x}^*) - \lambda_2 g_2(\boldsymbol{x}^*) - \cdots - \lambda_m g_m(\boldsymbol{x}^*) \\ \text{s.t. } & \lambda_1 \nabla g_1(\boldsymbol{x}^*) + \lambda_2 \nabla g_2(\boldsymbol{x}^*) + \cdots + \lambda_m \nabla g_m(\boldsymbol{x}^*) = \nabla f(\boldsymbol{x}^*) \\ & \lambda_i \geq 0 \text{ for each } i = 1, \ldots, m. \end{aligned}$$

Because the primal has a solution $\boldsymbol{h} = 0$, which leads to optimal value 0. By the duality theorem, the optimal value of the dual is also 0, implying that

$$-\lambda_1^* g_1(\boldsymbol{x}^*) - \lambda_2^* g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* g_m(\boldsymbol{x}^*) = 0.$$

Because $\lambda_i^* \geq 0$ and $g_i(\boldsymbol{x}^*) \leq 0$, $-\lambda_i^* g_i(\boldsymbol{x}^*) \geq 0$ for all $i = 1, \ldots, m$. Thus, it must be that

$$\lambda_i^* g_i(\boldsymbol{x}^*) = 0 \quad \text{for all } i = 1, \ldots, m. \qquad (\textit{complementary slackness})$$

Since the dual constraints must be satisfied, we see that a set of *necessary* conditions for $\boldsymbol{x}^*$ being a solution of the non-linear program is that there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\begin{gathered} \nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0, \\ g_i(\boldsymbol{x}^*) \leq 0 \quad \text{for all } i = 1, \ldots, m, \\ \lambda_i^* g_i(\boldsymbol{x}^*) = 0 \quad \text{for all } i = 1, \ldots, m. \end{gathered}$$

These conditions are sometimes called the **Langrange conditions**; the dual variables λ_1, ..., λ_m are also called **Lagrange multipliers**.

Since the Lagrange multipliers are just dual variables, they admit the same interpretation as what we discussed in the previous lecture.

Strictly speaking, what we obtained in the previous slide are the necessary conditions for $\boldsymbol{x}^*$ to be a solution of the *linear approximation* of the original non-linear program.

To see the only possible gap, note that the first condition can be written as

$$\nabla f(\boldsymbol{x}^*) = \lambda_1^* \nabla g_1(\boldsymbol{x}^*) + \lambda_2^* \nabla g_2(\boldsymbol{x}^*) + \cdots + \lambda_m^* \nabla g_m(\boldsymbol{x}^*);$$

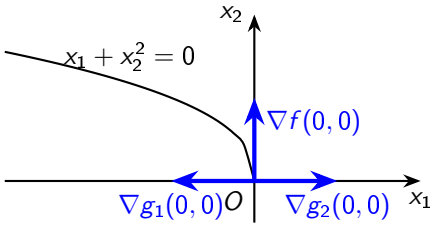
since $\lambda_i^* \geq 0$ for all $i = 1, \ldots, m$, the above equality means that $\nabla f(\boldsymbol{x}^*)$ is in the cone generated by $\{\nabla g_1(\boldsymbol{x}^*), \ldots, \nabla g_m(\boldsymbol{x}^*)\}$.

- As long as $\boldsymbol{x}^*$ is a solution, this is always possible for a linear program.
- However, the constraint set of non-linear programs may be ill-behaved, having "cusps" or pathological "turning points."
- Consequently, it can be that although $\boldsymbol{x}^*$ is indeed a solution to the non-linear program, $\nabla f(\boldsymbol{x}^*)$ is *not* in the cone generated by $\{\nabla g_1(\boldsymbol{x}^*), \ldots, \nabla g_m(\boldsymbol{x}^*)\}$.

Example: Consider the maximization problem

$$\max_{(x_1,x_2)\in\mathbb{R}^2} f(x_1,x_2) = x_2 \qquad \text{subject to} \quad g_1(x_1,x_2) = -x_1 \leq 0, \quad g_2(x_1,x_2) = x_1 + x_2^2 \leq 0.$$

The only point in $\mathbb{R}^2$ that satisfies the constraints is $(0,0)$; so $(0,0)$ must be the unique solution.



Now $\nabla f(0,0) = (0,1)$, $\nabla g_1(0,0) = (-1,0)$, and $\nabla g_2(0,0) = (1,0)$.

- Observe that $\nabla f(0,0)$ is not in the cone generated by $\{\nabla g_1(0,0), \nabla g_2(0,0)\}$
- Therefore we can't find $\lambda_1, \lambda_2 \geq 0$ such that $\nabla f(0,0) = \lambda_1 \nabla g_1(0,0) + \lambda_2 \nabla g_2(0,0)$
- Lagrange conditions are *not* necessary in that they're not satisfied by the unique solution

Necessary conditions

Thankfully, when the constraint functions are *all* affine (like in a linear program, although the objective function can be non-linear), we don't have to worry:

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a continuously differentiable function, and each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is an affine function. If $\boldsymbol{x}^* \in S$ and it is a solution to problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t. } g_i(\boldsymbol{x}) \leq 0 \text{ for } i = 1, \ldots, m,$$

then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\begin{gathered} \nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0, \\ g_i(\boldsymbol{x}^*) \leq 0 \quad \text{for all } i = 1, \ldots, m, \\ \lambda_i^* g_i(\boldsymbol{x}^*) = 0 \quad \text{for all } i = 1, \ldots, m. \end{gathered}$$

The condition $\nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0$ can also be obtained through an auxiliary function called the **Lagrangian**:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) - \lambda_1 g_1(\boldsymbol{x}) - \lambda_2 g_2(\boldsymbol{x}) - \cdots - \lambda_m g_m(\boldsymbol{x}).$$

If $\boldsymbol{x}^* \in S$ is a solution to the non-linear program, then there exists a $\boldsymbol{\lambda}^* \geq 0$ such that

$$\begin{aligned}
\frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_1} &= \frac{\partial f(\boldsymbol{x}^*)}{\partial x_1} - \lambda_1^* \frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_1} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_1} = 0, \\
\frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_2} &= \frac{\partial f(\boldsymbol{x}^*)}{\partial x_2} - \lambda_1^* \frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_2} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_2} = 0, \\
&\vdots \\
\frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_n} &= \frac{\partial f(\boldsymbol{x}^*)}{\partial x_n} - \lambda_1^* \frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_n} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_n} = 0.
\end{aligned}$$

Therefore, the solution $\boldsymbol{x}^*$ can be interpreted as a "critical point" of $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}^*)$, where the first-order conditions hold (but don't forget the constraints themselves, $g_i(\boldsymbol{x}^*) \leq 0$, and complementary slackness, $\lambda_i^* g_i(\boldsymbol{x}^*) = 0$).

If some of the constraint functions are not affine, as long as the set of gradients of constraint functions *binding at* $\boldsymbol{x}^*$ is linearly independent, we also have no trouble:

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ and $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$ be continuously differentiable functions. Suppose $\boldsymbol{x}^* \in S$ is a solution to the problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t. } g_i(\boldsymbol{x}) \leq 0 \text{ for } i = 1, \ldots, m,$$

and suppose the first $k \leq m$ constraints are binding, and the other $m - k$ constraints are slack. If the set of gradients of the binding constraints, $\{\nabla g_1(\boldsymbol{x}^*), \ldots, \nabla g_k(\boldsymbol{x}^*)\}$, is linearly independent, then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \geq 0$ such that

$$\begin{gathered} \nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0, \\ g_i(\boldsymbol{x}^*) \leq 0 \quad \text{for all } i = 1, \ldots, m, \\ \lambda_i^* g_i(\boldsymbol{x}^*) = 0 \quad \text{for all } i = 1, \ldots, m. \end{gathered}$$

The trickiness with the linear independence test is that we can't determine which constraints are binding until we have identified candidate solutions from the Lagrange conditions.

What we usually do instead is to test the linear independence of the set of gradients of *all* constraint functions, $\{\nabla g_1(\boldsymbol{x}), \ldots, \nabla g_m(\boldsymbol{x})\}$, *before* writing down the Lagrange conditions

- If the set $\{\nabla g_1(\boldsymbol{x}), \ldots, \nabla g_m(\boldsymbol{x})\}$ is linearly independent for *every* $\boldsymbol{x}$ in the constraint set, then we're completely safe.
- If not, things become more complicated, but in simple problems the number of points $\boldsymbol{y}$ in the constraint set such that $\{\nabla g_1(\boldsymbol{y}), \ldots, \nabla g_m(\boldsymbol{y})\}$ is linearly dependent is usually small.
- When there are a small number of constraints, we can check what happens when each of the subsets of the constraints bind (more on that in the next lecture).

Example: Consider the maximization problem

$$\max_{(x_1,x_2)\in\mathbb{R}^2} x_1^2 + x_2^2 \qquad \text{s.t.} \quad x_1 + x_2 - 1 \geq 0.$$

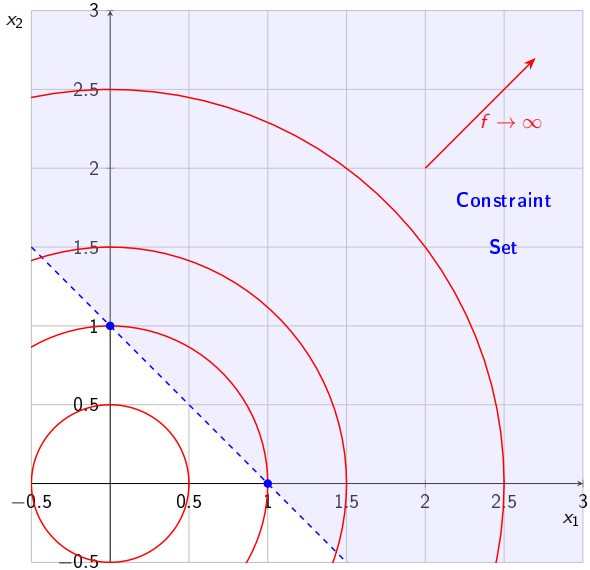
- To apply the theorem, we first need to write the constraint as $g(x_1, x_2) \equiv 1 - x_1 - x_2 \leq 0$.
- One way to see that the Lagrange conditions are necessary is that the lone constraint function is affine. One can also note that $\nabla g(x_1, x_2) = (1, 1) \neq (0, 0)$ for all $(x_1, x_2) \in \mathbb{R}^2$, and hence $\{\nabla g(x_1, x_2)\}$ is linearly independent for every (x_1, x_2) in the constraint set.
- The Lagrange conditions are

$$\begin{aligned} 2x_1 + \lambda &= 0 \\ 2x_2 + \lambda &= 0 \\ \lambda(1 - x_1 - x_2) &= 0 \\ \lambda &\geq 0 \\ 1 - x_1 - x_2 &\leq 0 \end{aligned}$$

Rule of thumb: The most convenient way of solving the Lagrange conditions is usually to start from complementary slackness.

- If $\lambda = 0$, then from the first two equalities we see that $x_1 = x_2 = 0$, violating $1 - x_1 - x_2 \leq 0$. Thus, $\lambda > 0$.
- Consequently, complementary slackness requires $x_1 + x_2 = 1$.
- From the first two equalities ($2x_1 + \lambda = 0$ and $2x_2 + \lambda = 0$), we see that $x_1 = x_2 = -\lambda/2$.
- Together with $x_1 + x_2 = 1$, we get $x_1 = x_2 = 1/2$ and $\lambda = -1$. But $\lambda > 0$, a contradiction.
- Thus, there is no point that satisfies the Lagrange conditions for this problem.
- Because the Lagrange conditions are necessary, that is, every solution must satisfy the Lagrange conditions, the maximization problem has no solution.

Graphically:



If we can use the extreme value theorem to conclude that a solution exists, we have a useful "cookbook" procedure for solving non-linear programs:

We first check if *all* constraint functions are affine. If so,

- Write down the Langrange conditions.
- Every $\boldsymbol{x}$ that (together with some $\boldsymbol{\lambda}$) satisfies the Lagrange conditions is a candidate solution.
- The candidates at which the value of the objective function are the largest are the solutions.

If not:

- Check whether the set of gradients of the constraint functions is linearly independent; identify the points at which it is not.
- Write down the Langrange conditions.
- Every $\boldsymbol{x}$ that violates the linear independence condition or that (together with some $\boldsymbol{\lambda}$) satisfies the Lagrange conditions is a candidate solution.
- The candidates at which the value of the objective function are the largest are the solutions.

Exercise: Solve the maximization problem

$$\max_{(x,y)\in\mathbb{R}^2} f(x,y) = x + y \qquad \text{s.t.} \quad g(x,y) = x^2 + y^2 - 2 \leq 0.$$

- By the extreme value theorem, the maximization problem has a solution.
- Since $\nabla g(x,y) = (2x, 2y) = (0,0)$ only if $(x,y) = (0,0)$, $\{\nabla g(x,y)\}$ is linearly independent for every (x,y) in the constraint set except for $(0,0)$.
- However, $(0,0)$ doesn't make the constraint bind, and therefore the set of gradients of the *binding* constraints is always linearly independent.
- Thus, the Lagrange conditions are necessary; i.e., every solution of this problem must satisfy the Lagrange conditions.

- The Lagrange conditions are

$$1 - 2\lambda x = 0 \tag{1}$$
$$1 - 2\lambda y = 0 \tag{2}$$
$$\lambda(x^2 + y^2 - 2) = 0 \tag{3}$$
$$x^2 + y^2 \leq 2 \tag{4}$$
$$\lambda \geq 0 \tag{5}$$

- If $\lambda = 0$ then neither (1) nor (2) are satisfied. Hence $\lambda > 0$ and, by the complementary slackness condition (3), $x^2 + y^2 = 2$.
- (1) and (2) then imply $x = y$; plugging it into $x^2 + y^2 = 2$, either $x = y = 1$ or $x = y = -1$.
- But if $x = y = -1$, (1) or (2) implies that $\lambda = -1/2$, a contradiction.
- Thus, only point that satisfies the Lagrange conditions is $(x^*, y^*, \lambda^*) = (1, 1, 1/2)$.
- Then since a solution exists, and every solution must satisfy the Lagrange conditions, the unique solution is $(x^*, y^*) = (1, 1)$.

Equality constraints

On slide 6, we mentioned that every constraint can be written as a "less than or equal to" constraint. Specifically,

- If the constraint looks like $g(\boldsymbol{x}) \leq h(\boldsymbol{x})$, we can write it as $g(\boldsymbol{x}) - h(\boldsymbol{x}) \leq 0$.
- $g_i(\boldsymbol{x}) \geq 0$ can be written as $-g_i(\boldsymbol{x}) \leq 0$.
- $g_i(\boldsymbol{x}) = 0$ can be replaced by two inequality constraints: $g_i(\boldsymbol{x}) \leq 0$ and $-g_i(\boldsymbol{x}) \leq 0$.

If all constraint functions are affine, everything works out smoothly. However, if not, we need to test linear independence of the set of gradients of the binding constraint functions.

- The linear independence of that set is sensitive to the representation of the constraint set.
- To see why, consider the problem $\max f(\boldsymbol{x})$ subject to $g(\boldsymbol{x}) = 0$.
- This can be written as $\max f(\boldsymbol{x})$ subject to $g(\boldsymbol{x}) \leq 0$ and $-g(\boldsymbol{x}) \leq 0$.
- However, both constraints bind, and the set of gradients $\{\nabla g(\boldsymbol{x}), -\nabla g(\boldsymbol{x})\}$ is linearly dependent, and therefore we can't use the theorem on slide 15.

Thankfully, the power of the duality theorem allows us to handle equality constraints easily.

The linear program on slide 8 now becomes, replacing $\leq$ by $=$,

$$\max_{\boldsymbol{h}} \nabla f(\boldsymbol{x}^*) \cdot \boldsymbol{h} \qquad \text{subject to} \quad \nabla g_i(\boldsymbol{x}^*) \cdot \boldsymbol{h} = -g_i(\boldsymbol{x}^*) \text{ for all } i = 1, \ldots, m.$$

Its dual is

$$\begin{aligned} &\min_{\boldsymbol{\lambda}} -\lambda_1 g_1(\boldsymbol{x}^*) - \lambda_2 g_2(\boldsymbol{x}^*) - \cdots - \lambda_m g_m(\boldsymbol{x}^*) \\ &\text{s.t. } \lambda_1 \nabla g_1(\boldsymbol{x}^*) + \lambda_2 \nabla g_2(\boldsymbol{x}^*) + \cdots + \lambda_m \nabla g_m(\boldsymbol{x}^*) = \nabla f(\boldsymbol{x}^*) \end{aligned}$$

The only difference is that the sign of the dual variables (Lagrange multipliers) are *unrestricted*.

- Because the constraints are equalities rather than inequalities, complementary slackness is no longer meaningful.

The discussion in the previous two slides gives rise to the following result:

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a continuously differentiable function. Suppose

- *either each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is an affine function,*
- *or each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is continuously differentiable, and the set of gradients of constraint functions $\{g_1(\boldsymbol{x}^*), \ldots, g_m(\boldsymbol{x}^*)\}$ is linearly independent.*

If $\boldsymbol{x}^* \in S$ and it is a solution to problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t. } g_i(\boldsymbol{x}) = 0 \text{ for } i = 1, \ldots, m,$$

Then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \in \mathbb{R}^m$ such that $g_i(\boldsymbol{x}^*) = 0$ for each $i = 1, \ldots, m$, and

$$\nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0.$$

When the constraints are equalities, there is an even closer analogy between the first-order conditions and the Lagrange conditions. To see this, consider again the Lagrangian:

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) - \lambda_1 g_1(\boldsymbol{x}) - \lambda_2 g_2(\boldsymbol{x}) - \cdots - \lambda_m g_m(\boldsymbol{x}).$$

If $\boldsymbol{x}^* \in S$ is a solution to the non-linear program with equality constraints, there exists $\boldsymbol{\lambda}^* \in \mathbb{R}^m$:

$$\begin{aligned}
\frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_1} &= \frac{\partial f(\boldsymbol{x}^*)}{\partial x_1} - \lambda_1^* \frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_1} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_1} = 0, \\
&\vdots \\
\frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial x_n} &= \frac{\partial f(\boldsymbol{x}^*)}{\partial x_n} - \lambda_1^* \frac{\partial g_1(\boldsymbol{x}^*)}{\partial x_n} - \cdots - \lambda_m^* \frac{\partial g_m(\boldsymbol{x}^*)}{\partial x_n} = 0, \\
\frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial \lambda_1} &= g_1(\boldsymbol{x}^*) = 0, \quad \cdots, \quad \frac{\partial \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)}{\partial \lambda_m} = g_m(\boldsymbol{x}^*) = 0.
\end{aligned}$$

Therefore, if $\boldsymbol{x}^*$ is a solution, $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*)$ *is* a critical point of $\mathcal{L}(\boldsymbol{x}, \boldsymbol{\lambda})$, where the first-order conditions hold.

Example: Consider the maximization problem

$$\max_{(x_1,x_2)\in\mathbb{R}^2} f(x_1,x_2) = -2x_1^2 - 3x_2^2 \qquad \text{s.t.} \quad g(x_1,x_2) = x_1 + x_2 - 1 = 0.$$

- We can either use the fact that g is affine, or to notice that $\nabla g(x_1,x_2) = (1,1) \neq (0,0)$ for all $(x_1,x_2) \in \mathbb{R}^2$, and therefore $\{\nabla g(x_1,x_2)\}$ is linearly independent for every (x,y) in the constraint set, to conclude that the Lagrange conditions are necessary.
- The Lagrange conditions are

$$\begin{aligned} -4x_1 + \lambda &= 0 \\ -6x_2 + \lambda &= 0 \\ x_1 + x_2 - 1 &= 0 \end{aligned}$$

- From the first two equations we get $\lambda = 4x_1 = 6x_2$, and therefore $x_1 = \frac{3}{2}x_2$.
- Plugging into the third equation (the equality constraint), we get the unique solution $(x_1^*, x_2^*, \lambda^*) = (3/5, 2/5, 12/5)$.
- Because the extreme value theorem fails to hold (why?), we can only claim that *if* this problem has a solution, then it must be $(x_1^*, x_2^*) = (3/5, 2/5)$.

A more general result for problems with both equality and inequality constraints:

Theorem

Let S be an open subset of $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be a continuously differentiable function. Suppose

- *either each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is an affine function,*
- *or each $g_j : S \to \mathbb{R}$, $j = 1, \ldots, m$, is continuously differentiable, and the set of gradients of constraint functions binding at $\boldsymbol{x}^*$ is linearly independent.*

If $\boldsymbol{x}^* \in S$ and it is a solution to problem

$$\max_{\boldsymbol{x} \in S} f(\boldsymbol{x}) \qquad \text{s.t. } g_i(\boldsymbol{x}) = 0 \text{ for } i = 1, \ldots, k, \text{ and } g_i(\boldsymbol{x}) \leq 0 \text{ for } i = k+1, \ldots, m,$$

Then there exists $\boldsymbol{\lambda}^* = (\lambda_1^*, \ldots, \lambda_m^*) \in \mathbb{R}^m$ such that $\lambda_i \geq 0$ for each $i = k+1, \ldots, m$, and

$$\nabla f(\boldsymbol{x}^*) - \lambda_1^* \nabla g_1(\boldsymbol{x}^*) - \lambda_2^* \nabla g_2(\boldsymbol{x}^*) - \cdots - \lambda_m^* \nabla g_m(\boldsymbol{x}^*) = 0,$$
$$g_i(\boldsymbol{x}^*) = 0 \text{ for all } i = 1, \ldots, k, \text{ and } g_i(\boldsymbol{x}^*) \leq 0 \text{ for all } i = k+1, \ldots, m,$$
$$\lambda_i^* g_i(\boldsymbol{x}^*) = 0 \quad \text{for all } i = k+1, \ldots, m.$$